

The Utility Problem

A transformer is a non-rotating electrical machine that allows electrical energy to be transferred from one circuit to another by electromagnetic induction. The transformer is an essential element in the electricity distribution industry. On one side, it is the step-up transformers that ensure the transmission and distribution of electrical energy from the point of generation to the consumer is efficient - the losses are disproportionately lower when transmitted at higher voltages (HV, MV). On the other side, the step-down transformers are bringing the voltages to lower levels which are safe for end use in households and businesses.

Each utility has many transformers in its portfolio (thousands, tens of thousands...), and this asset group presents a significant portion of the overall asset balance sheet. To manage this important portfolio of assets, utilities developed asset management techniques which range from simple to more advanced. Among more advanced techniques for transformer asset management are operational analyses, like, for example, Transformer Loading Analysis.

Transformer Loading Analysis fulfills an important role in Asset Management since it brings dynamic variables into consideration. During operation, for example, for safety reasons, it is important to know how the transformer is loaded over time. It is not beneficial for a transformer to be underloaded nor overloaded. If the transformer is overloaded for a long time, the lifetime of the transformer may be shortened or transformer failures may also occur, which will then negatively affect the reliability indicators (SAIDI and SAIFI), which are very important for the utility. If, on the other hand, the transformer is underutilized, it may be an inefficient use of investment funds, as a transformer with a higher capacity usually has a higher price. Also, such a transformer will have higher no-load losses, which again is inefficient.

The creation of the process for regular Transformer Loading Analysis often brings multiple benefits in the form of improved decision-making in the Asset Management area and overall grid operations management.

The Data Solutions and Methods

The Awesense Energy Transition Platform is ready for such analysis. With the Awesense Platform, it is very easy to analyze the transformer load either in the entire grid or only in selected parts of the grid. The unique integration of geospatial data (GIS) with time series data in the Awesense Platform enables this type of analysis.

The required datasets for this type of analysis are:

1. Asset Information - load loss (LL), no-load loss (NLL), Rated Power (kVA)
2. Loading data:
 - a. From the sensors /meter (if it exists) or,
 - b. Inferred from aggregated downstream consumption.

The approach to such analysis is significantly simplified. With the Awesense Platform, required data can be accessed in an easy manner since the needed datasets are part of the Awesense Energy Data Model (EDM) data layer.

To get the loading data for the transformer, the simplest and most accurate option is, of course, the one where this transformer is measured/monitored. This can be achieved with the *ts_data_source_select()* function. Then this measured data is compared with Asset Information to create the analysis. The asset information data required are transformer-rated power (units in kVA), no-load loss and load loss (units in W). In some cases, however, the transformer is not measured/monitored, so the real loading data is unavailable. In such cases, the transformer loading data can be inferred from the downstream load. The analyst would combine functions *grid_get_downstream()* and *ts_data_source_select()* to get the desired outcome.

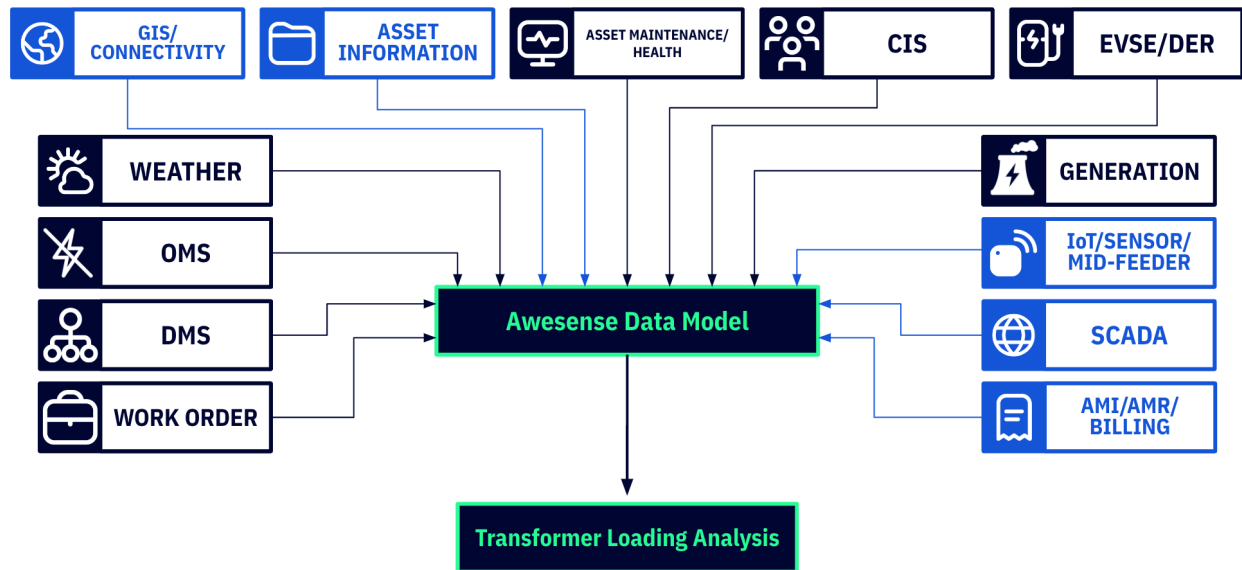
In the True Grid Intelligence (TGI) visualization environment, it is possible to display any transformer on a map or table for a more comprehensive evaluation - using segment analysis, it is possible to assign a calculated result to each transformer and to color-code transformers directly in the TGI visualization environment. Loading data can be fed into TGI for further analysis. In this way, it is possible to detect parts of the grid (or individual transformers) that are overloaded or underloaded.

All information about the selected transformer can also be displayed directly in the TGI visualization environment. There is detailed static information about the transformer, how the transformer is loaded over time, what % of the time it is overloaded or underloaded, a histogram with measured data along with the ideal load profile of such a transformer. The report is generated automatically for each transformer. It is also possible to set up alerts for any transformer parameters using grid alerts directly in TGI. Any measured quantities and any threshold values can be set. For example, exceeding the nominal value of the transformer. All settings are made by selecting from a list of parameters, and everything is very simple and flexible. If the data are integrated with near real-time, a real transformer monitoring tool is available.

All of these functionalities provide a detailed overview of transformer usage and status, which then helps to optimize the operation and maintenance of transformers and therefore save significant resources.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. This use case is implemented in TGI as part of the Transformer Loading Assessment/Analysis functionality. Consult [instructions](#) about how to access this use case in TGI. If you do not have access to a server where the use case functionality is available, contact [Awesense](#) to see a demo of it.
2. A Jupyter notebook implementation can also be provided upon request.